

Timeline

Eleven weeks from kickoff. Dates assume Week 1 starts Monday Sep 14. Week 11 lands on Thanksgiving week (Nov 23–27), so confirm the club's Demo Day date and shift the last two weeks if needed.

At a glance

Week	Dates	Focus	What exists by the end
1	Sep 14–18	Setup + teams	Teams assigned, repo ready, everyone can build
2–3	Sep 21–Oct 2	Core systems	Basic panel parsing, finger and panel tracking, guidance, camera UI, each working alone
4–5	Oct 5–16	Working prototype	A finger is guided to a detected button on a real appliance
6–7	Oct 19–30	Reliable core	Guidance works across realistic conditions with clear errors; outside testers can use it
8–9	Nov 2–13	Product expansion	The highest-value features the core can support
10	Nov 16–20	Testing + stabilization	Demo build works repeatedly on real appliances
11	Nov 23–27	Freeze + Demo Day	Feature freeze, polish, rehearsal, backup plan

Gates

Four decisions that don't slip.

- End of Week 1: everyone builds and runs the project on a device. If not, we fix that before anything else.
 - End of Week 2: interfaces (SurfaceMap, TrackingFrame, GuidanceState) and coordinate convention frozen. Guidance model chosen from the bake-off.
 - End of Week 5: the vertical slice works on a real appliance. If it doesn't, Weeks 6–7 are spent making it work, and Weeks 8–9 shrink.
 - Start of Week 8: expansion features chosen based on what guidance can actually do. If guidance is still unreliable, we improve it instead.
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Week 1 — Project Setup + Team Formation (Sep 14–18)

Goal: everyone is ready to start development.

Officers, before Monday:

- Base Xcode project with one Swift package per team, all compiling empty
- Repo with .gitignore, PR template, branch protection on main, CI running build and tests
- Project board with Week 2–3 tasks written per team, small and specific
- The Interfaces section of the Architecture tab finalized as a first draft

Monday kickoff with the whole club:

- Explain the problem, the core interaction, the architecture, and the four teams
- Each lead introduces their team in a few minutes: what it owns, what the first milestone is, who it suits
- Share the doc afterward. Juniors submit team preferences by Wednesday.

Rest of the week:

- Teams assigned by Thursday. Every junior knows their team and their lead.
- Every developer clones, builds, and runs the app on a physical device. Signing and provisioning issues get solved in Slack, not alone.
- Every team member sends five photos of appliance panels they have access to. That's the first test set.

- Device inventory: who has an iPhone 15 Pro or newer. Anyone without one is paired with someone who does.

Checkpoint: every developer knows what their team owns and can begin working.

Weeks 2–3 — Core Systems (Sep 21–Oct 2)

Goal: get each part of the core interaction working independently, against mocks.

Surface Understanding

- Detect an appliance interface in a photo
- Identify basic buttons and labels
- Produce a basic SurfaceMap
- Hand-label the test set

Tracking / Spatial

- Track the index fingertip
- Track the panel as the phone moves
- Report finger position relative to a known target

Feedback / Guidance

- Run the guidance model bake-off (Week 2) and pick one
- Prototype proximity haptics
- Prototype directional audio
- Define success and lost-tracking feedback with Design

Product / UI

- Live camera experience
- Basic target selection and guidance flow
- Mocks for all three interfaces, shipped by Monday Sep 21 so everyone has something to build against
- First integration build Friday Oct 2, even if it's mostly mocks

Design

- State-by-state feel spec, agreed with Feedback and Product
- Camera acquisition and target selection flows

Checkpoint: we can separately show where a button is, where the finger is, where it needs to move, and how that movement is communicated.

Weeks 4–5 — Guide a Finger to a Detected Button (Oct 5–16)

Goal: connect everything into the first complete Contour prototype.

See appliance → identify buttons → select target → track panel and finger → calculate movement → guide user → reach button

- Focus is integration between Surface → Tracking → Feedback → Product. Mocks come out one at a time.
- Expect coordinate-system, latency, lost-tracking, and interface problems. Fix them as they appear; write down what was decided.
- Each team keeps a short list of what breaks when the real thing replaces the mock.
- UI can still be rough. Speech can still be robotic. It has to work.
- Internal demo Friday Oct 16: someone on the team who didn't build it tries it on a real microwave.

Checkpoint: point at a real appliance, request a button, and successfully guide a finger to it.

Weeks 6–7 — Make Button Guidance Reliable and Usable (Oct 19–30)

Goal: turn the prototype into a dependable core experience.

Improve, in roughly this order:

- Panel recognition across different appliances
- Fingertip and panel tracking stability
- Camera positioning instructions: how the user gets the panel in frame without seeing the screen
- Target selection and confirmation
- Guidance clarity: does the user know which way to move?
- Confidence and error handling
- Success and failure states
- VoiceOver and accessibility
- Overall speed

Testing

- Start testing with people outside the team. Reach out to CIDI in Week 6 for testers in Week 7.
- Test on at least five different appliances and two lighting conditions.
- Track what fails and why. That list drives Week 8.

Checkpoint: someone unfamiliar with the implementation can use Contour to reach a requested button without developer help.

Weeks 8–9 — Expand Around Working Guidance (Nov 2–13)

Goal: add depth without weakening the core.

Choose from these based on progress. Do not commit to all of them.

- Explore Panel
- Read Display
- Low-Vision Mode
- Onboarding and guidance tutorial

- Saved Appliances
- Simple Task Mode

Rule: if guidance is still unreliable, improve it instead of adding features. Core teams keep improving recognition, tracking, guidance, and performance regardless.

Each expansion feature gets one owner and a one-week cut line. If it isn't demoable by Friday Nov 13, it's out.

Checkpoint: the core is still reliable and the added features make Contour meaningfully more useful.

Week 10 — Real-World Testing + Demo Stabilization (Nov 16–20)

Goal: make the Demo Day build predictable.

Test repeatedly across:

- Different appliance panels
- Lighting and glare
- Phone positions and distances
- Full start-to-finish interactions, many times in a row

Fix in this order: incorrect guidance → crashes → tracking or recognition failures → speed → confusing UX → polish

Also finalize:

- The demo appliance and the demo task
- The spectator visualization for the projector: what the audience sees while the user feels
- The fallback demo if the live one fails

Checkpoint: the planned demo can be repeated successfully without developer intervention.

Week 11 — Feature Freeze + Demo Day (Nov 23–27)

Goal: stop building and make what exists excellent.

- Feature freeze. Only important bugs get fixed.
- Finish UI and demo polish.
- Rehearse the full presentation at least twice, once with the real appliance and projector.
- Test devices, appliance, projection, and spectator view in the actual room if possible.
- Rehearse the fallback plan.

Final demo: point at appliance → Contour reads the buttons → request a button → live guidance → reach the correct button → clear confirmation. Ideally the person holding the phone is from the audience and has their eyes closed.

Checkpoint: reliable product, clear story, confident demonstration.